

## TONGLIAO, China

WMO: 541350

Lat: 43.5954N

Lon: 122.2603E

Elev: 591

StdP: 14.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.6       | -3.8 | -23.0                         | 1.6 | -0.3 | -19.4 | 1.9 | 2.6  | 22.9                  | 16.7 | 18.8 | 15.9 | 5.0                   | 290  | 0.519 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.3                   | 91.6            | 70.2 | 88.9 | 69.9 | 86.5 | 68.9 | 77.0                | 84.6 | 75.3 | 82.5 | 73.6 | 81.0 | 8.8                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.8                            | 133.2 | 81.5 | 73.1 | 125.6 | 80.2 | 71.3 | 118.2 | 78.4 | 41.1          | 84.5 | 39.4 | 82.2 | 37.8 | 80.7 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |     |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |     |
|-------------------|------|------|----------------------------|------|-----|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-----|
| 1%                |      | 2.5% | 5%                         | Mean |     | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |     |
| Min               | Max  | Min  | Max                        | Min  | Max | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g) | (h)                | (i)                                                | (j)       | (k)   | (l)        | (m)   | (n)        | (o)   | (p)        |     |
| 21.9              | 18.2 | 15.7 | -11.5                      | 96.9 | 5.5 | 4.1                | -15.5                                              | 99.8      | -18.7 | 102.2      | -21.7 | 104.5      | -25.7 | 107.5      |     |
|                   |      |      | -12.3                      | 79.4 | 5.2 | 2.3                | -16.1                                              | 81.1      | -19.1 | 82.4       | -22.1 | 83.7       | -25.9 | 85.4       |     |
|                   |      |      | DB                         |      |     |                    |                                                    |           |       |            |       |            |       |            |     |
|                   |      |      | WB                         |      |     |                    |                                                    |           |       |            |       |            |       |            |     |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 46.1   | 10.1  | 18.9  | 32.6  | 49.6  | 62.9  | 72.2  | 76.7  | 73.8  | 63.6  | 48.0  | 28.8  | 13.9  |
|                                                                   | DBStd       | 24.60  | 7.61  | 8.92  | 9.60  | 8.55  | 7.25  | 5.64  | 4.12  | 4.62  | 6.35  | 8.49  | 10.29 | 7.97  |
|                                                                   | HDD50       | 4663   | 1237  | 871   | 545   | 110   | 3     | 0     | 0     | 0     | 2     | 138   | 639   | 1118  |
|                                                                   | HDD65       | 7899   | 1702  | 1291  | 1005  | 467   | 126   | 9     | 0     | 3     | 98    | 528   | 1087  | 1583  |
|                                                                   | CDD50       | 3226   | 0     | 0     | 5     | 97    | 404   | 666   | 829   | 738   | 409   | 76    | 2     | 0     |
|                                                                   | CDD65       | 986    | 0     | 0     | 0     | 3     | 62    | 225   | 364   | 275   | 55    | 2     | 0     | 0     |
|                                                                   | CDH74       | 8735   | 0     | 0     | 1     | 69    | 706   | 2081  | 3189  | 2166  | 500   | 23    | 0     | 0     |
|                                                                   | CDH80       | 2983   | 0     | 0     | 0     | 12    | 234   | 808   | 1171  | 663   | 93    | 2     | 0     | 0     |
|                                                                   | WSAvg       | 7.2    | 6.5   | 7.2   | 8.4   | 9.2   | 8.7   | 7.1   | 6.5   | 5.9   | 6.4   | 7.1   | 7.0   | 6.5   |
| Precipitation                                                     | PrecAvg     | 14.7   | 0.0   | 0.1   | 0.2   | 0.6   | 1.1   | 2.7   | 4.5   | 3.2   | 1.3   | 0.7   | 0.2   | 0.1   |
|                                                                   | PrecMax     | 21.6   | 0.2   | 0.4   | 1.6   | 3.3   | 2.7   | 5.0   | 11.7  | 8.9   | 3.2   | 3.4   | 0.7   | 0.9   |
|                                                                   | PrecMin     | 7.9    | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 1.9   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                   | PrecStd     | 3.0    | 0.1   | 0.1   | 0.3   | 0.7   | 0.7   | 1.4   | 2.1   | 2.1   | 0.8   | 0.8   | 0.2   | 0.2   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 36.0  | 48.0  | 67.4  | 81.1  | 91.7  | 96.1  | 95.2  | 92.3  | 86.3  | 77.4  | 60.8  | 42.2  |
|                                                                   |             | MCWB   | 28.3  | 34.5  | 48.0  | 55.6  | 61.2  | 67.6  | 74.9  | 72.8  | 67.3  | 58.6  | 46.1  | 34.7  |
|                                                                   | 2%          | DB     | 30.0  | 42.5  | 58.6  | 75.0  | 86.1  | 91.3  | 91.6  | 88.8  | 82.7  | 71.4  | 54.0  | 34.4  |
|                                                                   |             | MCWB   | 23.5  | 32.0  | 42.3  | 52.7  | 60.9  | 66.9  | 73.5  | 72.3  | 65.0  | 55.8  | 43.3  | 27.9  |
|                                                                   | 5%          | DB     | 26.2  | 38.2  | 53.0  | 70.2  | 81.5  | 88.0  | 89.0  | 86.2  | 79.5  | 66.2  | 48.8  | 30.2  |
|                                                                   |             | MCWB   | 20.7  | 29.6  | 38.8  | 50.5  | 59.0  | 65.9  | 72.6  | 71.0  | 62.7  | 52.5  | 39.7  | 24.5  |
|                                                                   | 10%         | DB     | 22.6  | 33.7  | 48.3  | 65.3  | 77.2  | 84.8  | 86.4  | 83.7  | 76.5  | 62.1  | 43.8  | 26.1  |
|                                                                   |             | MCWB   | 18.0  | 26.5  | 35.9  | 48.0  | 56.6  | 65.4  | 71.7  | 70.0  | 61.2  | 50.3  | 35.7  | 21.3  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 28.8  | 36.7  | 48.8  | 58.5  | 66.2  | 73.5  | 80.2  | 78.7  | 72.3  | 62.4  | 49.2  | 35.0  |
|                                                                   |             | MCDB   | 35.6  | 45.7  | 64.5  | 74.8  | 83.1  | 86.0  | 90.4  | 86.9  | 81.2  | 69.0  | 55.3  | 41.5  |
|                                                                   | 2%          | WB     | 23.8  | 32.7  | 43.1  | 55.1  | 63.2  | 71.3  | 77.9  | 76.6  | 69.0  | 58.6  | 44.3  | 28.2  |
|                                                                   |             | MCDB   | 29.6  | 41.9  | 56.7  | 70.4  | 79.6  | 82.7  | 86.6  | 83.7  | 77.2  | 67.6  | 53.7  | 33.9  |
|                                                                   | 5%          | WB     | 20.9  | 29.8  | 39.7  | 51.9  | 61.1  | 69.7  | 76.2  | 75.0  | 66.7  | 55.4  | 39.9  | 24.5  |
|                                                                   |             | MCDB   | 26.2  | 38.1  | 51.9  | 66.8  | 76.0  | 80.5  | 83.6  | 82.2  | 74.7  | 63.7  | 48.4  | 30.1  |
|                                                                   | 10%         | WB     | 18.0  | 26.5  | 36.7  | 49.3  | 59.0  | 68.2  | 74.6  | 73.3  | 64.4  | 52.0  | 35.9  | 21.3  |
|                                                                   |             | MCDB   | 22.5  | 33.7  | 47.6  | 63.7  | 73.8  | 78.4  | 81.7  | 79.8  | 72.7  | 60.6  | 43.5  | 26.0  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.6  | 19.1  | 19.8  | 21.3  | 21.0  | 18.9  | 16.3  | 17.2  | 21.3  | 20.2  | 17.6  | 16.9  |
|                                                                   |             | MCDBR  | 22.5  | 24.1  | 27.1  | 29.2  | 28.3  | 25.0  | 21.0  | 20.5  | 25.4  | 26.2  | 23.8  | 21.2  |
|                                                                   |             | MCWBR  | 17.7  | 16.8  | 15.7  | 14.4  | 11.3  | 9.0   | 7.6   | 7.8   | 10.7  | 13.4  | 15.3  | 16.1  |
|                                                                   | 5% WB       | MCDBR  | 22.3  | 23.8  | 25.6  | 25.4  | 23.7  | 18.9  | 16.9  | 16.2  | 19.8  | 22.1  | 23.3  | 21.0  |
|                                                                   |             | MCWBR  | 17.7  | 16.9  | 15.3  | 13.4  | 10.8  | 8.1   | 7.9   | 8.0   | 10.1  | 13.6  | 15.8  | 16.2  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.342 | 0.401 | 0.503 | 0.576 | 0.643 | 0.600 | 0.538 | 0.471 | 0.438 | 0.357 | 0.316 |
|                                                                   | taud        |        | 2.190 | 2.114 | 1.995 | 1.786 | 1.683 | 1.637 | 1.776 | 1.903 | 1.982 | 1.959 | 2.094 | 2.142 |
|                                                                   | Ebn at Noon |        | 255   | 264   | 261   | 242   | 228   | 213   | 221   | 231   | 236   | 223   | 227   | 234   |
|                                                                   | Edh at Noon |        | 30    | 38    | 49    | 66    | 75    | 79    | 68    | 58    | 49    | 44    | 32    | 28    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 725   | 1041  | 1415  | 1725  | 1899  | 1798  | 1743  | 1667  | 1467  | 1058  | 724   | 599   |
|                                                                   | RadStd      |        | 55    | 68    | 82    | 115   | 125   | 145   | 141   | 88    | 69    | 70    | 51    | 48    |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | -1.75  | N/A   | N/A     | N/A   | N/A | N/A         | +121  | N/A   |       |
| (47) | Regional (5 neighbors) | N/A   | N/A     | -1.75  | N/A   | N/A     | N/A   | N/A | N/A         | +88   | N/A   |       |

Nomenclature: See separate page